

Experiments with cpVAE on string order in spin models

Generating data

$$H = \sum_{j=1}^{L-1} \alpha \mathbf{S}_j \mathbf{S}_{j+1} + \beta \left(\mathbf{S}_j \mathbf{S}_{j+1} \right)^2$$

The general spin-1 model. We use $\alpha=\cos(\theta)$ and $\beta=\sin(\theta)$.

- $\theta=0.0$: Heisenberg model
- $\arctan(\theta)=1/3$: AKLT model

Benchmark of the code for generating data

Exactly-solvable AKLT model

$$H^{\text{AKLT}} = \sum_i P_{i,i+1}^{S=2} = \sum_i \frac{1}{2} S_i S_{i+1} + \frac{1}{6} (S_i S_{i+1})^2 + \frac{1}{3}.$$

	Exact	Code
Ground energy per site*	0.0 (-0.6324555)	-0.632456 (scaled)
Entropy	log 2 ~ 0.69	0.693147
String order	4/9 ~ 0.44	0.444444
Correlation	Exponential decay	Exponential decay

Ground energy

Due to the constant difference of the exact-solvable form (where the ground energy is 0.0) and coded AKLT Hamiltonian, the true ground energy (per site) is

$$-2/3 \cos(\theta), \theta = \arctan(1/3) = -0.6324555$$

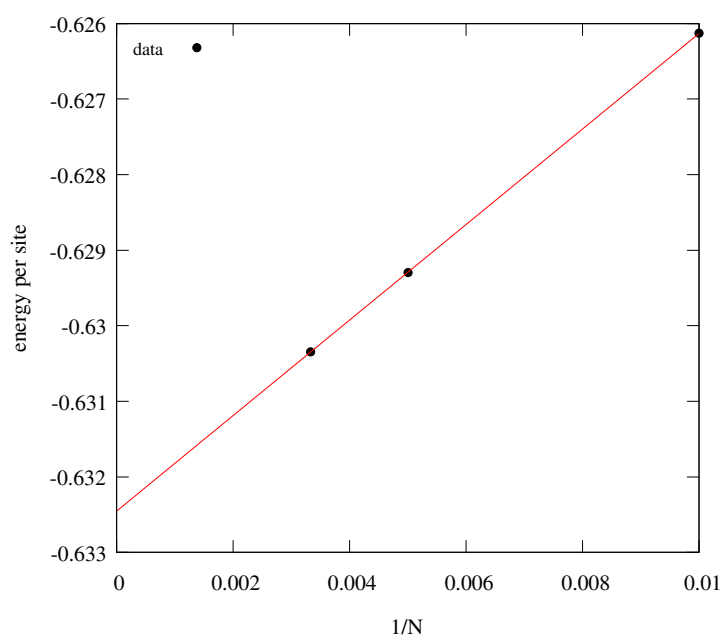
Computational conditions:

- N = 200 (number of spin), OBC but N = 100,200,300 is used for scaling
- Correlation functions: Let i0 is the centering site (i0=N/2) and calculate until i0+r=N
- String order correlation function ($\alpha=z$)

$$O_{\text{string}}^\alpha = \lim_{|i-j| \rightarrow \infty} \left\langle -S_i^\alpha \left(\prod_{k=i+1}^{j-1} e^{i\pi S_k^\alpha} \right) S_j^\alpha \right\rangle$$

- Correlation function ($\alpha=x,y,z$)

$$\mathcal{O}_d^{\text{FM},\alpha} = \mathcal{O}_d(S^\alpha) = \langle \psi | S_i^\alpha S_{i+d}^\alpha | \psi \rangle$$



System size scaling of ground energy per site

Benchmark of the code for generating data

Sampling

We calculate the expectation values using sampling generated from the resulting MPS through DMRG, to know how big samples is enough to recover the true expectation value.

We calculate the z-axis spin and the string correlator, and compare the results from the normally calculated value in MPS term (true) and the samplings (relative error). For simplicity, we use the AKLT for comparison.

$$H = \sum_{i=1}^{N-1} \cos \theta S_i S_{i+1} + \sin \theta (S_i S_{i+1})^2$$

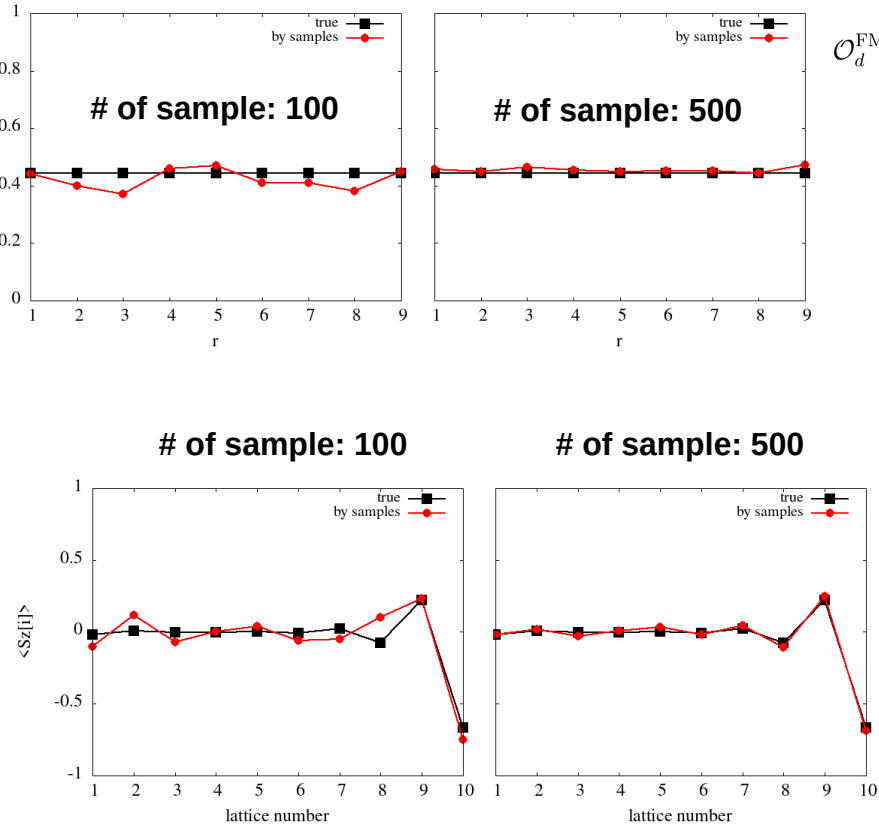
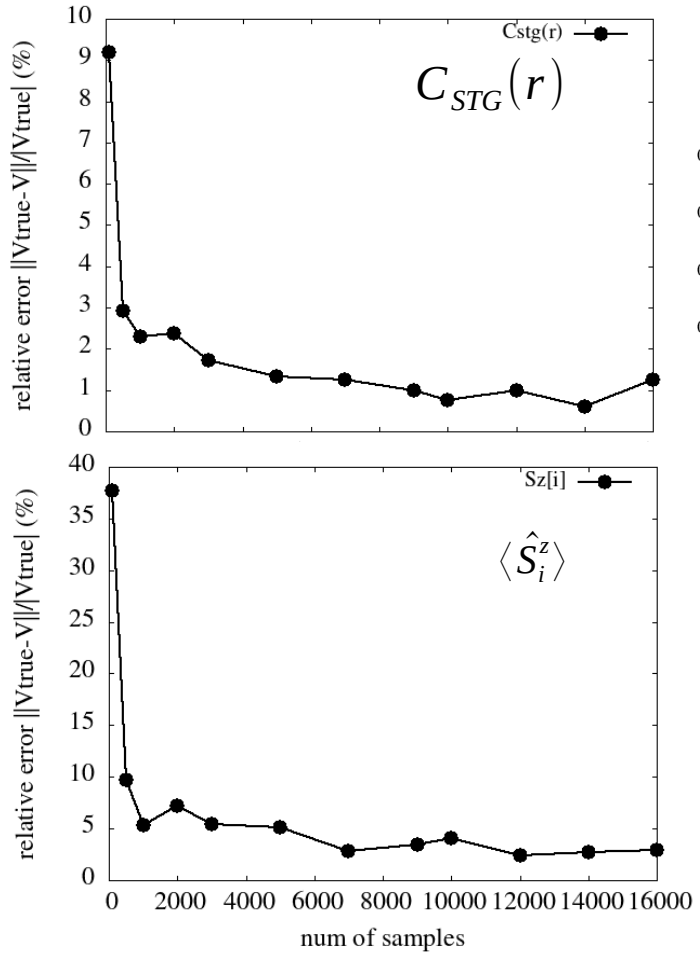
Computational conditions:

- **N = 10** (number of spin), $\theta = \arctan(1/3)$ [equivalent to AKLT]
- String order correlation function ($\alpha=z$)

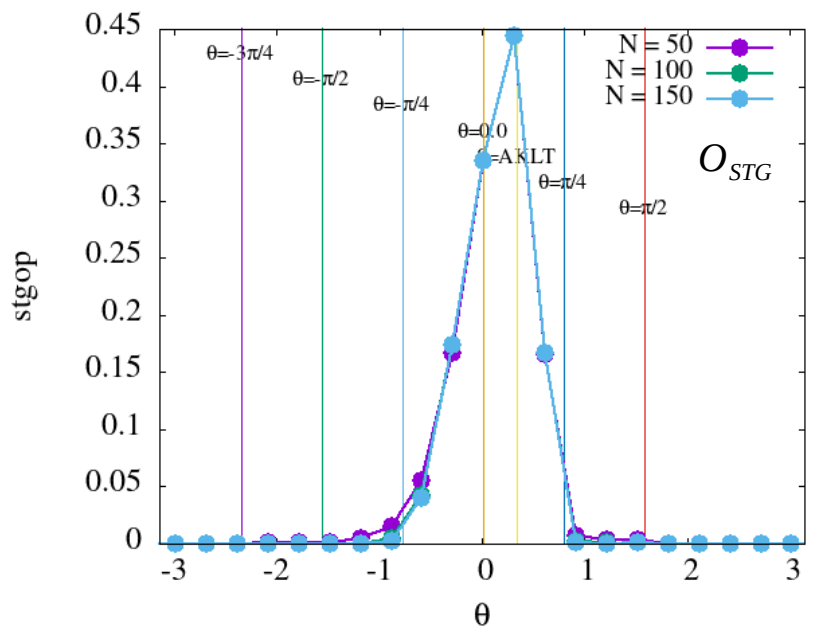
$$O_{\text{string}}^\alpha = \lim_{|i-j| \rightarrow \infty} \left\langle -S_i^\alpha \left(\prod_{k=i+1}^{j-1} e^{i\pi S_k^\alpha} \right) S_j^\alpha \right\rangle$$

- Correlation function ($\alpha=z$)

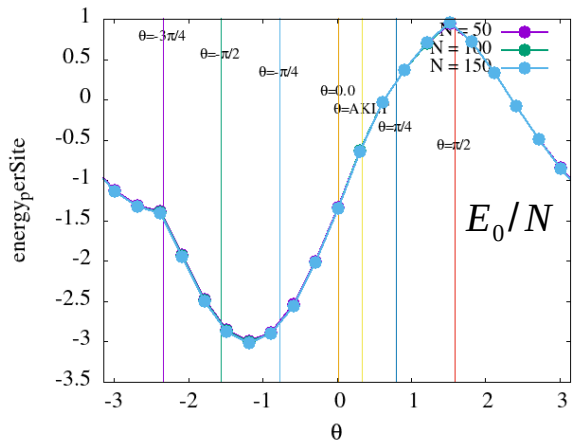
$$O_d^{\text{FM},\alpha} = O_d(S^\alpha) = \langle \psi | S_i^\alpha S_{i+d}^\alpha | \psi \rangle$$



Varying θ



$$C(r)=C(i=1,j=i+r)$$
$$O=C(r=r_{max})=C(i=1,j=N)$$



$$H=\sum_{i=1}^{N-1} \cos \theta S_i S_{i+1} + \sin \theta (S_i S_{i+1})^2$$

- Computational conditions:
- **N = 50,100,150** (number of spin)
 - OBC
 - Maxbonddim = 100
 - max_sweep = 50 (stop before if energy_tol=1e-8)

Ref. Bachelor Thesis
Johannes Gutenberg University Mainz

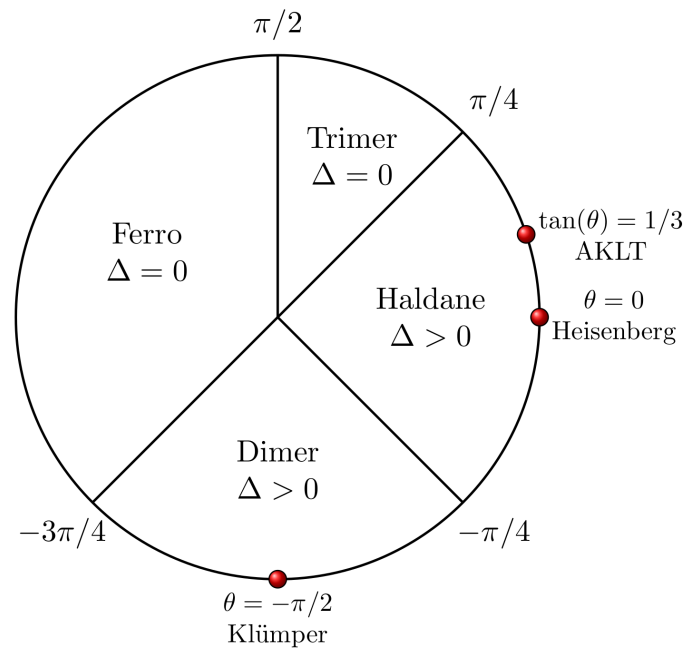
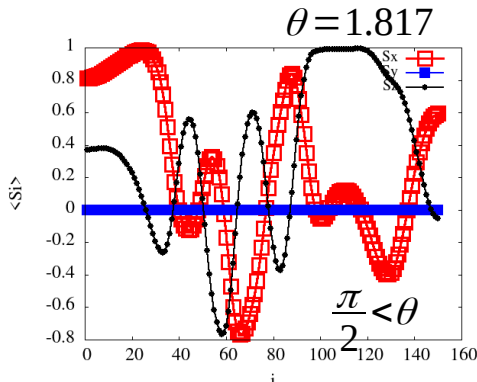


Figure 21: Phase chart for different values of θ . Δ represents the energy gap.

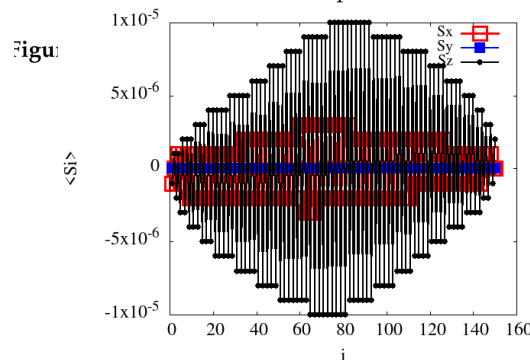
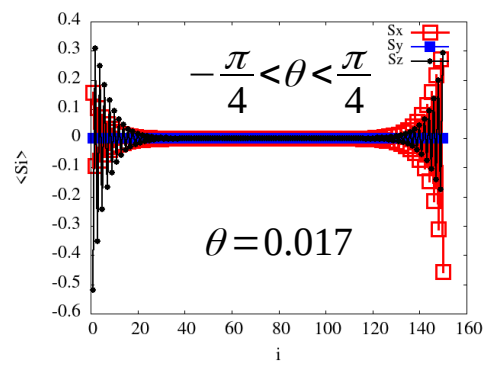
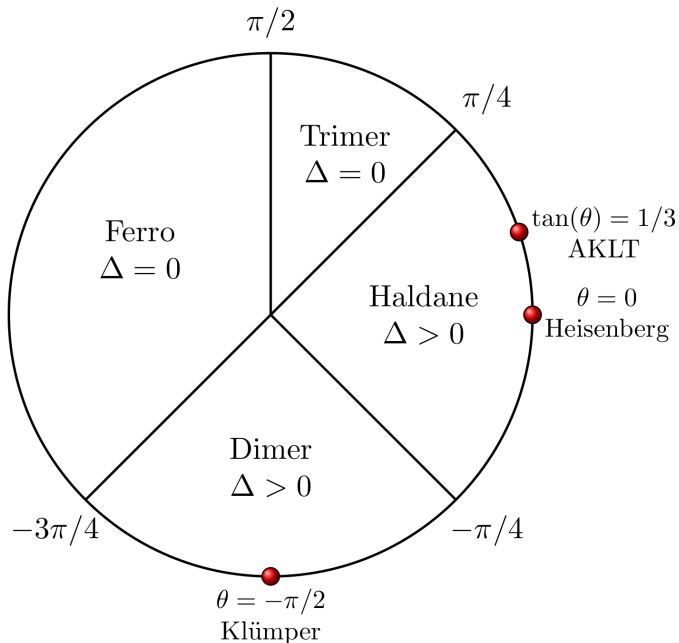
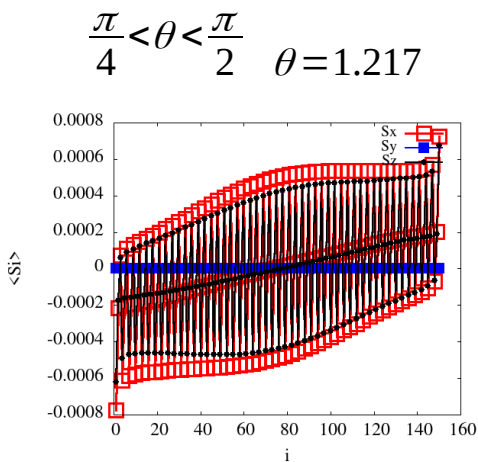
Varying θ , $N = 150$



$$H = \sum_{i=1}^{N-1} \cos \theta S_i S_{i+1} + \sin \theta (S_i S_{i+1})^2$$

- Computational conditions:
- **N = 150** (number of spin)
 - OBC
 - Maxbonddim = 100
 - max_sweep = 50 (stop before if energy_tol=1e-8)

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resents the energy gap.

$\theta = -1.483$